Top FCNC search at FCC-ee

Higgs performance meeting

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1. Motivation
2. Top-FCNC appearance at FCC-ee
3. Analysis strategy
   signal selection and Background estimation
4. Results
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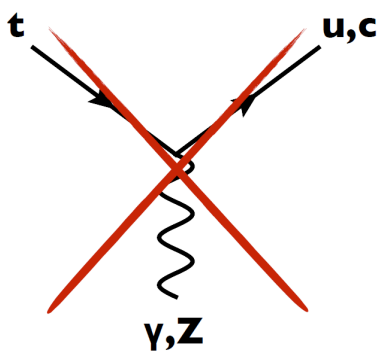
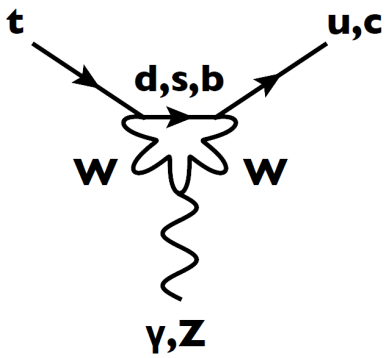
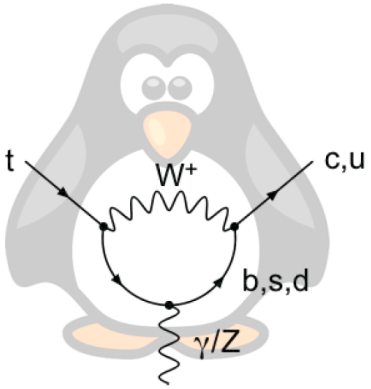
5. Summary and Conclusion

FCNC theory insights and Motivation

- In the SM, top FCNC is forbidden at tree level and highly suppressed at one-loop corrections due to GIM.
- $\mathcal{B}r(t \to qV) < 10^{-12}$ ($V = H, \gamma, Z, g$) in the SM, but can be enhanced up to 10^{-5} in some BSMs.
- Any observation of top FCNC would be a clear signal of new physics beyond the SM.
- FCC-ee offers ideal environment to study FCNC due to reduction of QCD background as well as high luminosity.

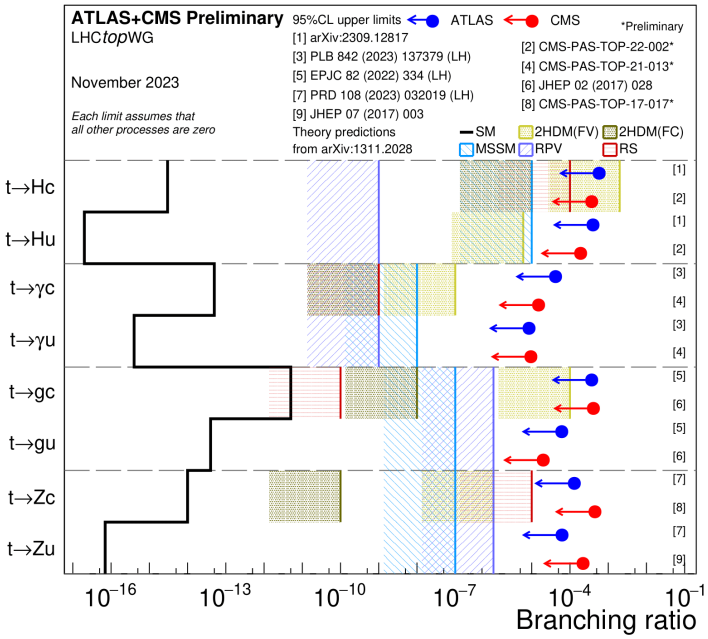


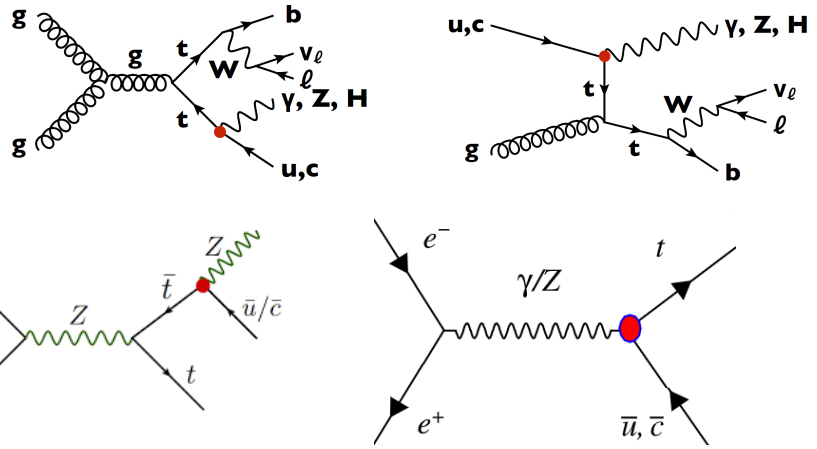




Experimental Limits

- LHC experiments have been searching for Top FCNC.
- So far, no clear evidence for the presence of top quark FCNC interactions has been observed and upper limits are set on the anomalous couplings and top quark branching.



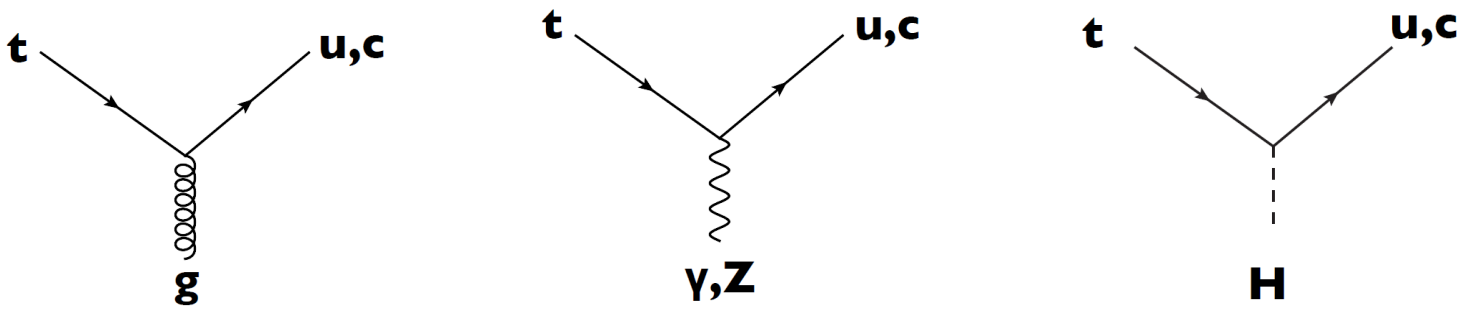


- OPAL & **600 pb**$^{-1}$ data collected at **189-209 GeV** & leptonic and the hadronic decay modes of the W boson & the Durham algorithm [*Physics Letters B 521 (2001)* 181-194]($Br(t \to zq) = 13.7\%$).
- L3 & **the total integrated luminosity of 634 pb**$^{-1}$ & $\sqrt{s}$ =**189 to 209 GeV** & Durham algorithm [*Physics Letters B 549 (2002) 290-300*] ($Br(t \to \gamma q) = 4.1\%$ and $Br(t \to Zq) = 13.7\%$).

Effective Framework

The most general effective Lagrangian describing the FCNC interactions can be parameterized as:

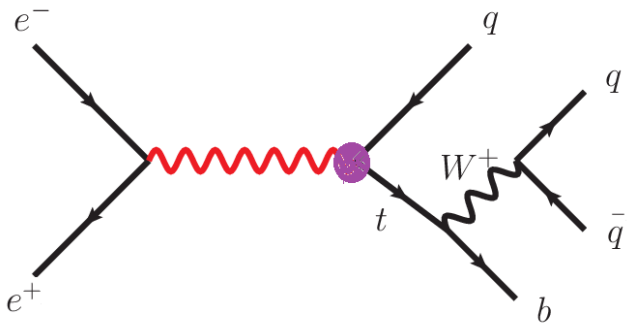
$$\mathcal{L}_{\text{FCNC}}^{tqX} = \sum_{q=u,c} \Bigg[\frac{g_s}{2m_t} \bar{q} \lambda^a \sigma^{\mu\nu} \left(\zeta_{qt}^L P^L + \zeta_{qt}^R P^R \right) t G_{\mu\nu}^a - \frac{1}{\sqrt{2}} \bar{q} \left(\eta_{qt}^L P^L + \eta_{qt}^R P^R \right) t H$$
$$+ \frac{g_W}{4 c_W m_Z} \bar{q} \sigma^{\mu\nu} \left(\kappa_{qt}^L P_L + \kappa_{qt}^R P_R \right) t Z_{\mu\nu} - \frac{g_W}{2 c_W} \bar{q} \gamma^\mu \left(X_{qt}^L P_L + X_{qt}^R P_R \right) t Z_\mu$$
$$+ \frac{e}{2m_t} \bar{q} \sigma^{\mu\nu} \left(\lambda_{qt}^L P_L + \lambda_{qt}^R P_R \right) t A_{\mu\nu} \Bigg] + \text{h.c.}\,.$$

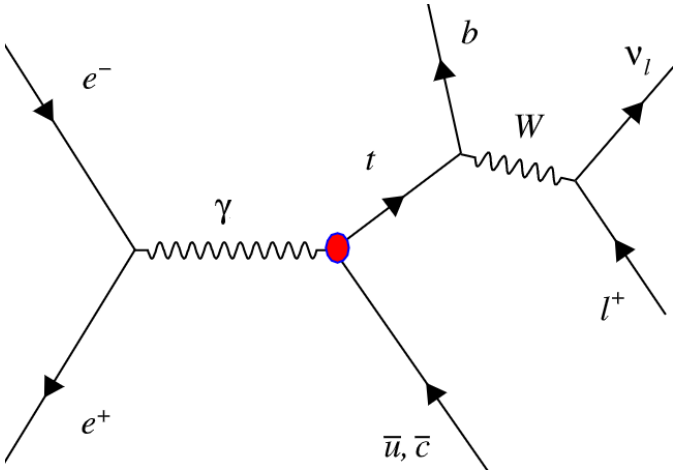


$\zeta_{qt}^L = \zeta_{qt}^R$ $\eta_{qt}^L = \eta_{qt}^R$ $\kappa_{qt}^L = \kappa_{qt}^R$ $X_{qt}^L = X_{qt}^R$ $\lambda_{qt}^L = \lambda_{qt}^R$

FCNC search at FCC-ee

- At the FCCee, FCNC tqV interaction can lead to single top production
- This analysis is performed at the center of mass energy 240 and 365 GeV.
- The distinctive signature of the signal is:
  - A top quark
  - A light energetic jet





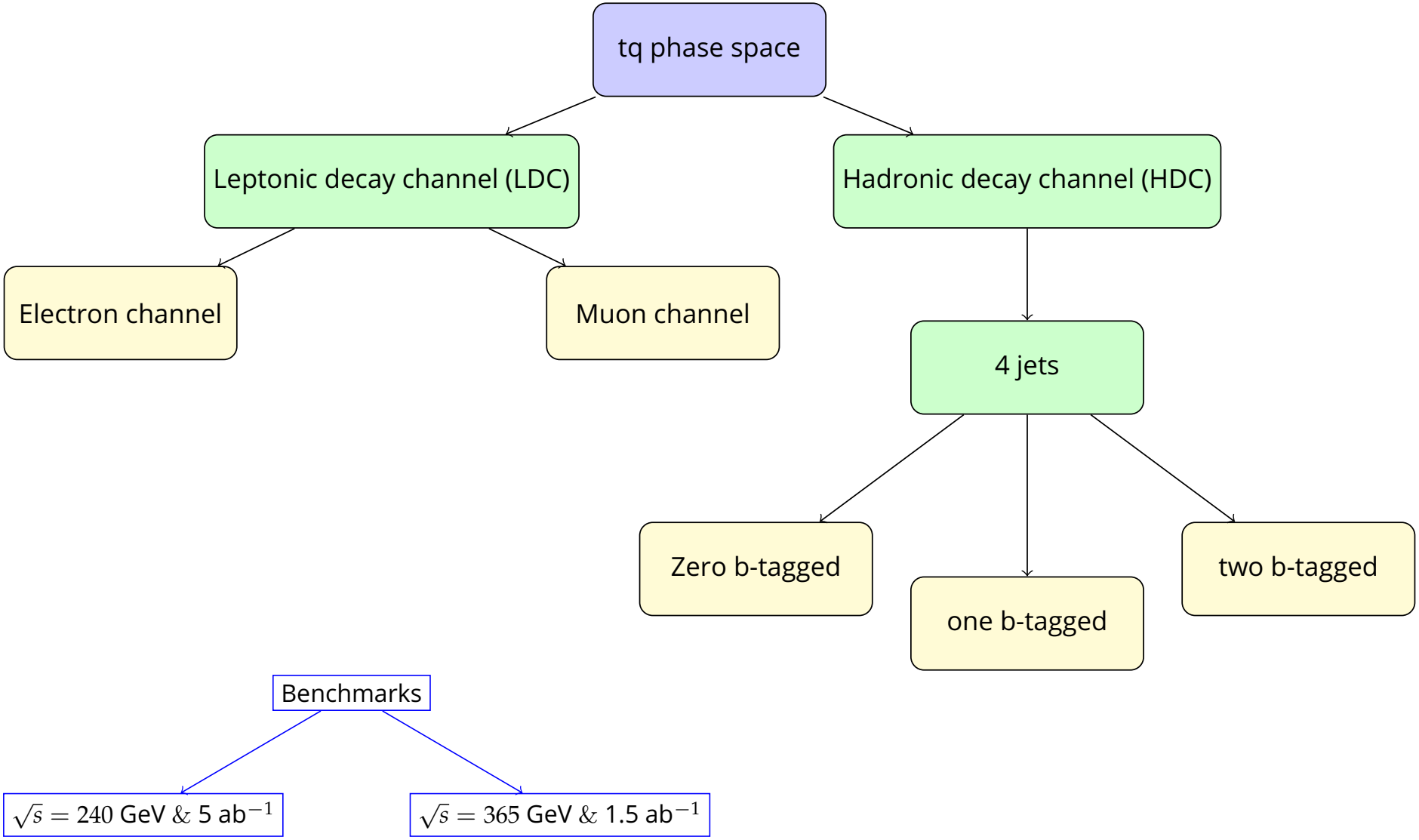
Signal signature at HDC

- 4 exclusive jets:
  - one or two of jets are tagged as b-jet
  - one energetic light jet

Signal signature at LDC

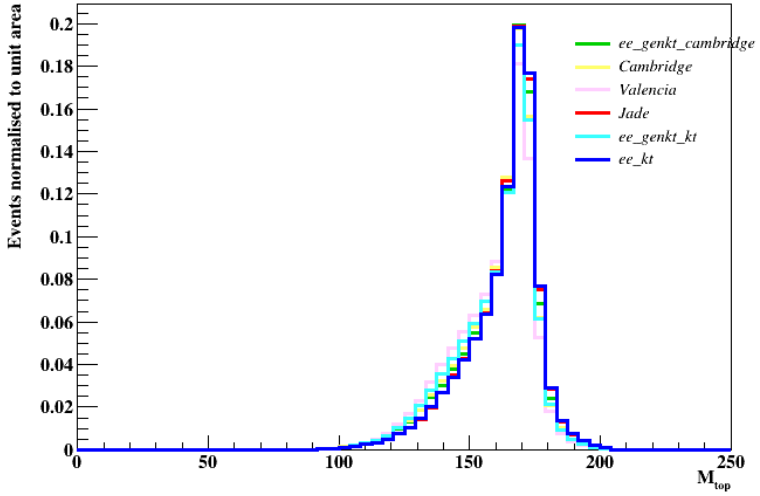
- One lepton
- one neutrino
- exactly 2 exclusive jets:
  - one jet is tagged as b-jet
  - one energetic light jet

Analysis categorisation

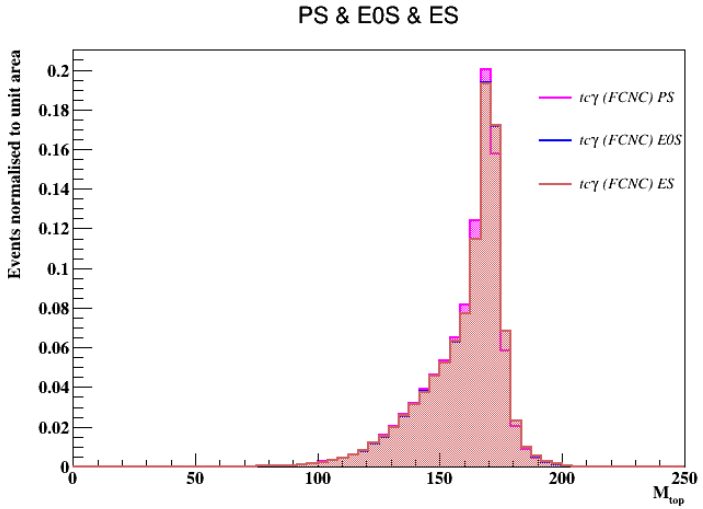


Jet clustering algorithms selection

Top mass reconstruction for different jet algorithms:



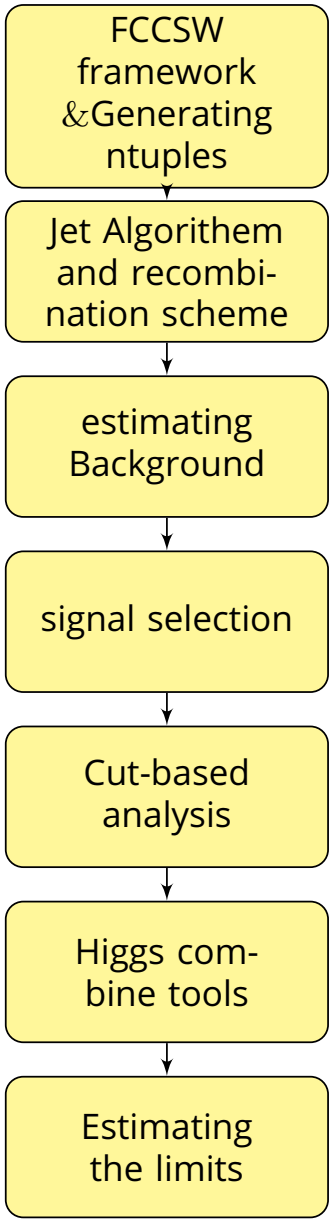
Reconstructed top mass distribution for three different schemes, for $tc\gamma$ coupling.



Detailed study on jet algorithm has been done in studying Top Anomalous Couplings in FCC-ee ($t\bar{t}$ production):

Jet Definition = Durham & E-scheme

Analysis overview



Backgrounds:

Process	σ[fb] $\sqrt{s} = 240$GeV	σ[fb] $\sqrt{s} = 365$GeV
ZH	202	117
WW	16439	10717
ZZ	1360	643
$e^-e^+q\bar{q}$	1637	620
single top	0.01	0.2
$tbjj$	-	6
$t\bar{t}$	-	800

preselection cuts:

- Lepton and jets momentum are required to have: $p > 15$GeV
- Polar angle of lepton and jets are required to be: $\theta > 13$ degree
- The angular separation between lepton and jets are required to satisfy: $\Delta R > 0.5$
- charged lepton veto (HDC)
- Missing energy is grater than 15 GeV (LDC)
- Missing energy is less than 20 GeV (HDC)

All MC files are generated in the FCCSW (https://github.com/HEP-FCC/FCCAnalyses) framework with $DelphesPythia8_EDM4HEP$ and IDEA Delphes Card

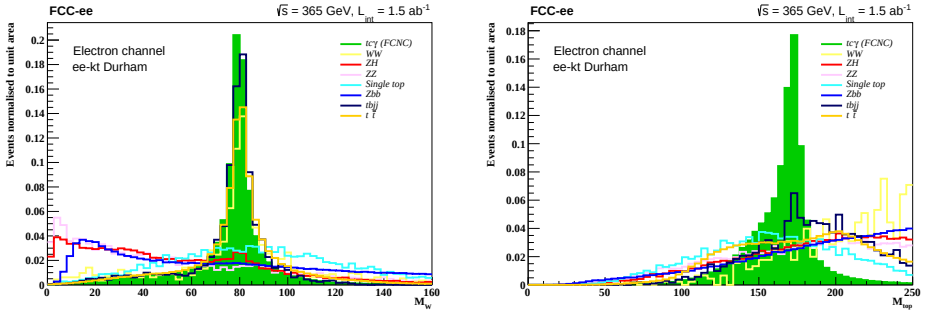
kinematic top reconstruction

- **Leptonic decay channel:**

$$M_T^2 = ((P_l + P_\nu) + P_b)^2,$$

with

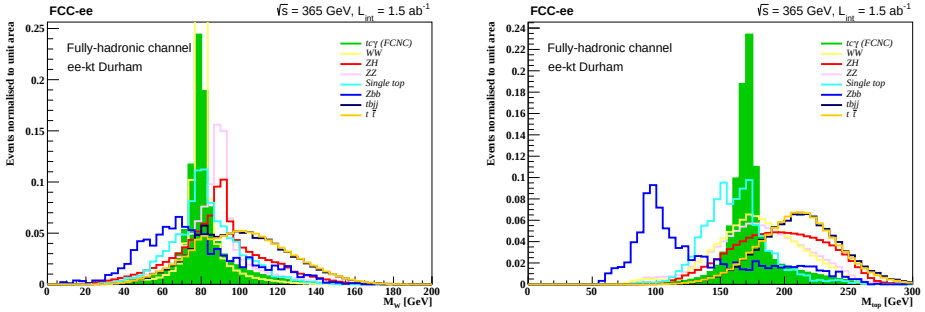
$$M_W^2 = (P_l + P_\nu)^2$$



- **Hadronic decay channel:** **Chi-square**

minimisation is performed to minimize the distance between jets invariant mass combinations and W/top mass:

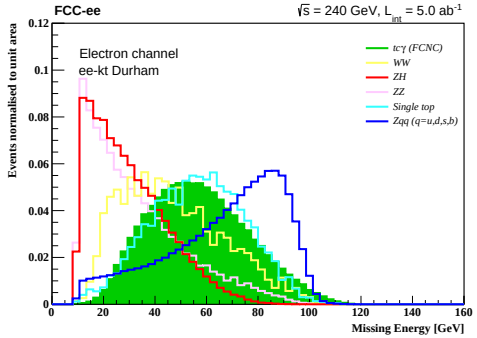
$$\chi^2 = (M_W - M_{jnjm}^{\text{reco}})^2 + (M_{\text{top}} - M_{jnjmj_l}^{\text{reco}})^2,$$

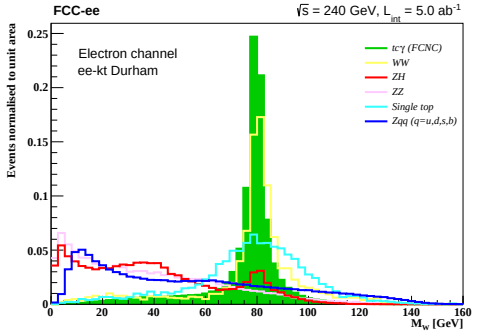


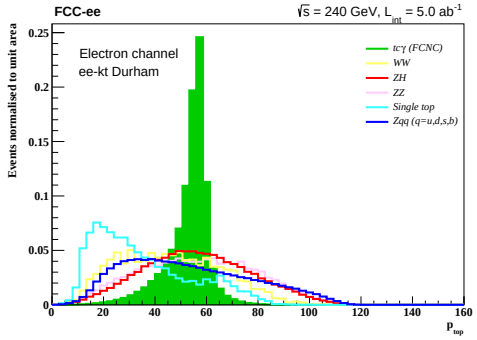
Distinctive variables-LDC

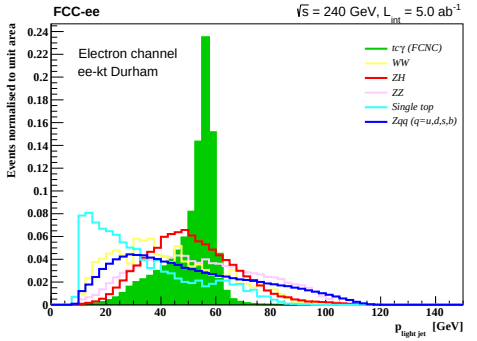
Optimal cuts on individual variables increase our sensitivity to the signal. The variables which have the best possible discrimination power between the signal and background processes:

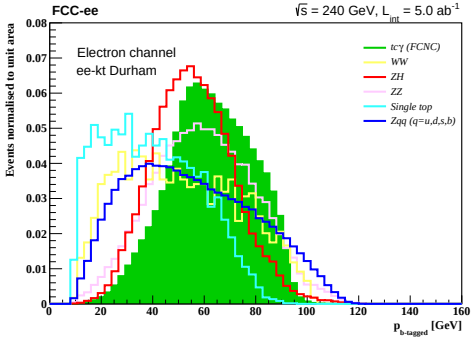
1. Top mass M_{top}
2. W boson mass M_{W}
3. Momentum of b-jet p_{bjet}
4. Momentum of light jet p_{lightjet}
5. Momentum of leptonp_{ℓ}
6. missing energy ME
7. Angular sepration between electron and b-jet $\Delta R(\ell, b_{\text{jet}})$
8. $E_{\text{bjet}}/(E_{\text{bjet}} + E_{\text{lightjet}})$.
9. scalar sum of momentum of all outgoing particles H_T

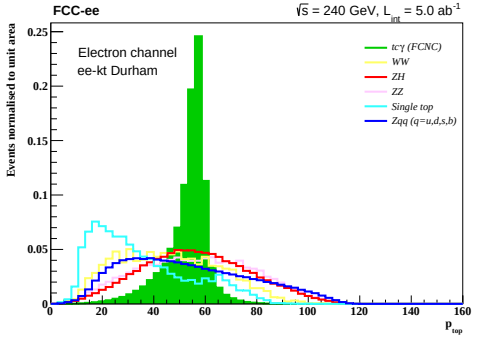


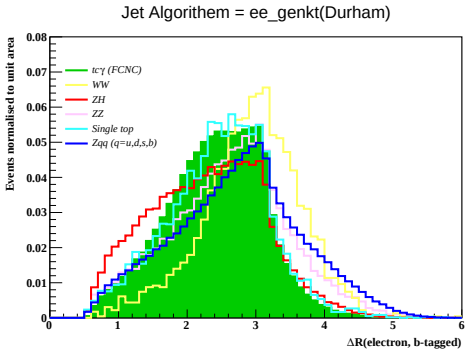












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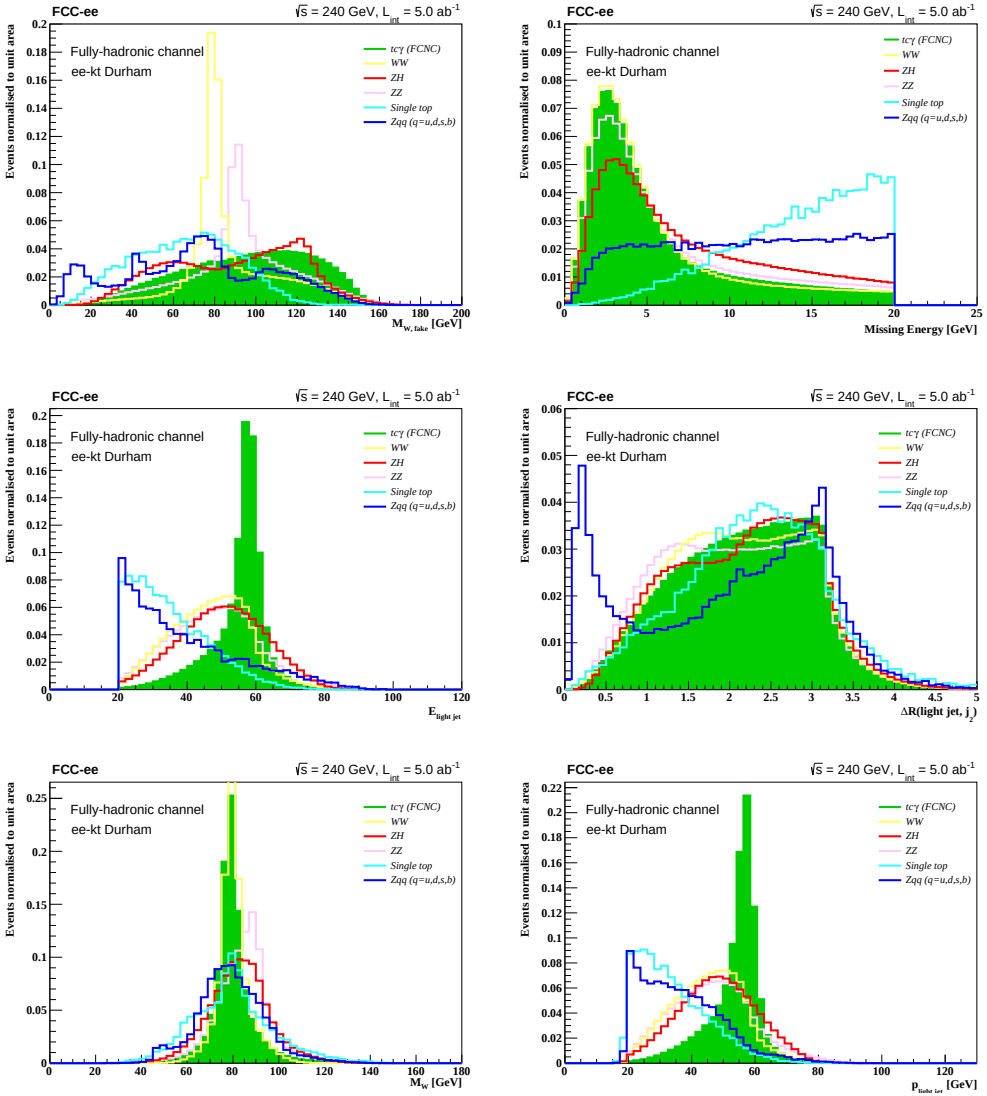


1. Top mass M_{top}
2. W boson mass M_{W}
3. **The reconstructed invariant mass of the fake W boson: $M_{\mathbf{W}} < 70$ GeV or $M_{\mathbf{W}} > 100$ GeV.**
4. Momentum of b-jet p_{bjet}
5. Momentum of light jet p_{lightjet}
6. Momentum of leptonp_{ℓ}
7. missing energy ME
8. Angular separation between jets $\Delta R(j_i, j_r)$
9. $E_{\text{bjet}}/(E_{\text{bjet}} + E_{\text{lightjet}})$.
10. scalar sum of momentum of all outgoing particles H_T
11. The ratio of the total momentum of the b-jet to the light jet $\frac{p_{\text{light-jet}}}{p_{\text{b-tagged}}}$

Cut flow tables for different signal scenarios

$\sqrt{s}$ = 240 GeV	$tc\gamma$	tcZ	ZH	WW	ZZ	eebb	eeqq (q=uds)	single top
Cross section	14.327	34.132	201.868	16438.5	1358.99	455.7	1635.9	0.010753
Preselection	3.1542	7.5154	1.1953	3.75948	2.76677	19.251	0.155146	0.0003208
Final selection	1.3686	3.2622	0.00847	0.254797	0.01127	0.1807	0.001439	3.2259×10^{-7}

240 GeV (LDC)

$\sqrt{s}$ = 365 GeV	$tc\gamma$	tcZ	ZH	WW	ZZ	$t\bar{t}$	eebb	single top	single top ($tbjj$)
Cross section	25.958	50.9517	117.3	10716.5	642.8	800.0	615.2	0.227	5.98
Preselection	5.2107	10.2299	0.6240	2.55696	1.21039	39.7574	25.0175	0.006297	0.2159
Final selection	1.9599	4.3406	0.00410	0.00857	0.00809	1.10752	0.004332	2.276×10^{-6}	0.006947

365 GeV (LDC)

$\sqrt{s}$ = 240 GeV	$tc\gamma$	tcZ	ZH	WW	ZZ	eebb	eeqq (q=uds)	single top
Cross section (fb)	28.39	67.44	201.868	16438.5	1358.99	455.7	1635.9	0.01075
Preselection	22.571	38.651	83.452	4285.272	400.682	0.2511	1.3863	0.00082
Final selection	6.609	11.320	8.361	257.030	34.193	0.00018	0.0027	4.83×10^{-7}

240 GeV (HDC)

$\sqrt{s}$ = 365 GeV	$tc\gamma$	tcZ	ZH	WW	ZZ	$t\bar{t}$	eebb	single top	$tbjj$
Cross section (fb)	52.39	104.0	117.3	10716.5	642.8	800.0	615.2	0.227	5.98
Preselection	39.152	76.435	32.779	2137.639	0.2038	271.443	0.1626	0.0034	2.973
Final selection	14.499	28.318	2.194	52.307	5.775	5.689	8.66×10^{-5}	~ 0	0.055

365 G

Results

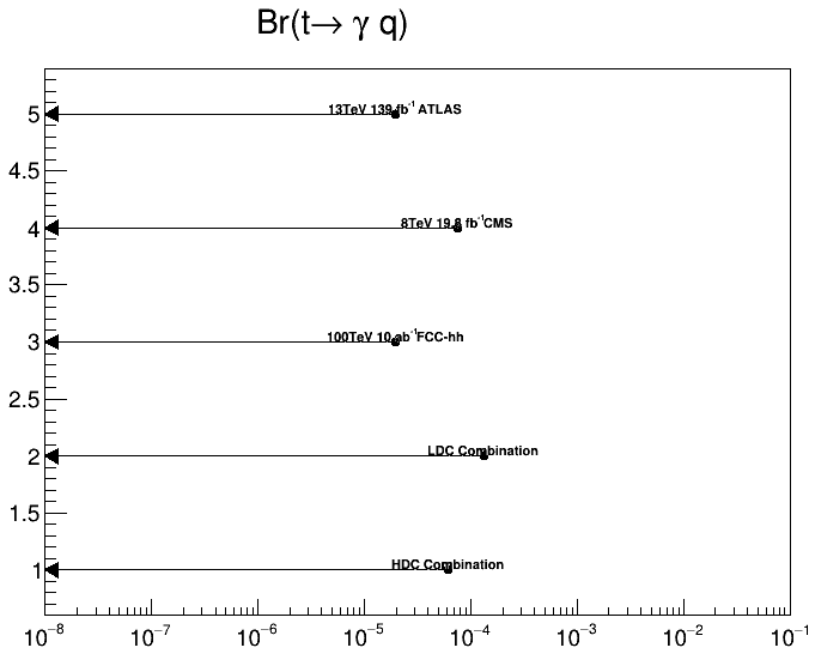
95% CL upper limts on the anomalous FCNC couplings and consequently on the branching ratios are estimated by using Higgs Combine tools.

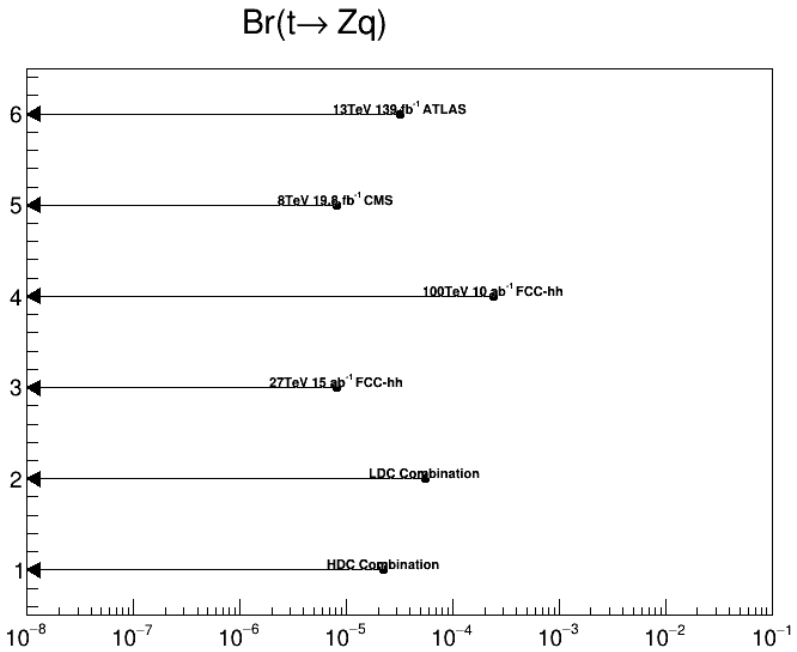
$\sqrt{s} = 240$ GeV & $\mathcal{L} = 5$ ab^{-1}	$Br(t \to c\gamma)$	$Br(t \to cZ)$
Electron channel	9.60×10^{-5}	3.53×10^{-5}
Muon channel	6.29×10^{-5}	2.31×10^{-5}
Full hadronic decay channel	1.98×10^{-4}	1.01×10^{-4}

$\sqrt{s} = 365$ GeV & $\mathcal{L} = 1.5$ ab^{-1}	$Br(t \to c\gamma)$	$Br(t \to cZ)$
Electron channel	1.65×10^{-4}	6.96×10^{-5}
Muon channel	1.47×10^{-4}	6.19×10^{-5}
Full hadronic decay channel	8.45×10^{-5}	3.70×10^{-5}

	$Br(t \to c\gamma)$	$Br(t \to cZ)$
Energy & channel statistical combination	1.99×10^{-5}	8.19×10^{-6}
Energy statistical combination	7.65×10^{-5}	3.28×10^{-5}

Comparison with current and future experiments





Summary

1. The sensitivity of the FCC-ee to top FCNC interaction is probed by searching tq production.
2. This study has been done in the official FCCAnalysis framework.
3. both leptonic and hadronic decay of the top qurak is considered.
4. We presented the upper limits on the branching ration for the leptonic (electron & muon) channel and hadronic decay channel of single-top quark production via FCNC at two FCC-ee benchmarks ($\sqrt{s} = 240$ GeV at 5 ab^{-1} and $\sqrt{s} = 365$ GeV at 1.5 ab^{-1}).
5. We show the statistical combination will improve the sensitivity to $tq\gamma$ and $tc\gamma$ couplings.

Thank
you